

**BCA (Semester-II) Examination, 2022**

**(Session 2021-23)**

**DISCRETE MATHEMATICS**

**| Paper Code : BCA-201 |**

Time : Three Hours

[Maximum Marks : 80]

**Note:** Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. Define a Hasse-Diagram for the Poset  $(P(S), \subseteq)$  where  $P(S)$  is power set on set  $S = \{1, 2, 3\}$ .

2. Define partitions of the set S and find out the partitions of  $X = \{1, 2, 3\}$ .

3. If  $f(x) = 3x - 5$  and  $f[g(x)] = 2x$  then find  $g(x)$ .

4. Show that set of all divisors of 12 form a Lattice.

5. Define equivalence relation with an example.

6. Is "Congruence Modulo n" is an equivalence relation? Explain it.

7. In a class of 25 students 12 have taken mathematics, 8 have taken mathematics but not biology. Find the number of students who have taken mathematics and biology and those who have taken biology but not mathematics.

8. Define Group with an example.

9. Let  $A = \{1, 2, 3, 4, 5\}$ ,  $B = \{3, 4, 5, 6\}$ ,  $C = \{x \in \mathbb{Z} : x \text{ is even}\}$ .

Find:

- (a)  $A - B$
- (b)  $B \cap A$
- (c)  $C - B$
- (d)  $(A \cap B) \cup C$

10. Write short notes on any two of the following:

- (a) Monoid
- (b) Partition of sets
- (c) Equivalence relation
- (d) Lattice

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